Total No. of Questions : 8]

SEAT No. :

PD-4896

[Total No. of Pages : 2

[6404]-453
B.E. (Mechanical)
(Honors in Electric Vehicles)
E-VEHICLE STANDARDS, CHARGING AND SAFETY
(2019 Pattern) (Semester - VIII) (302036MJ)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) *Answer Q. 1 or Q. 2, Q. 3 or Q. 4., Q.5 or Q.6 and Q.7 or Q.8.*
2) *Neat diagrams must be drawn whenever necessary.*
3) *Figures to the right side indicate full marks.*
4) *Use of Calculator is allowed.*
5) *Assume suitable data, if necessary.*

Q1) a) Explain Role of EV aggregators in EVGI framework. **[9]**

b) Write a note on Electric Vehicle Charging and grid integration standards. **[8]**

OR

Q2) a) Explain Control and communication infrastructure in EV charging? **[9]**

b) Discuss the potential challenges and opportunities associated with widespread adoption of electric vehicles in the context of grid integration. **[8]**

Q3) a) Describe Testing procedures for EV batteries in brief. **[9]**

b) Explain the role of regulatory standards in establishing safety requirements for function batteries in electric vehicles. **[9]**

OR

Q4) a) Explain specific Requirements for L and N Category Electric Power Train Vehicles. **[9]**

b) What is the blunt indentation test and how does it contribute to the evaluation of battery safety? **[9]**

P.T.O.

Q5) a) What are the key parameters measured during electrochemical and thermal characterization of batteries? **[9]**

b) Explain : **[8]**

i) Battery Aging in an Electric Vehicle (EV)

ii) Calorimetry

OR

Q6) a) List and explain several reasons for Lithium-ion battery failure. **[9]**

b) Explain the role of regulatory standards and industry guidelines in ensuring the safety of battery technology in electric vehicles. **[8]**

Q7) a) Explain in detail about EV charging infrastructures. Also explain categories of charging stations. **[9]**

b) How does location planning and land allocation impact the accessibility and efficiency of EV charging networks? **[9]**

OR

Q8) a) Explain Charging Methods & Power Ratings of Electric Vehicle. **[9]**

b) How does EV charging infrastructure contribute to energy diversification and efficiency within the power sector? **[9]**

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